

MOCK EXAMINATION — TRY BEFORE YOU BOOK

GCSE Computer Science

Mock Test & Mark Scheme

Time allowed	25 minutes
Total marks	34 marks
Number of questions	27
Topics covered	Computational Thinking, Data Representation, Programming, Networks & Security

Student name: _____ **Date:** _____

Instructions

- Answer every question in the space provided on your own paper.
- Write your name and today's date at the top of your answer sheet.
- The marks for each question are shown in brackets, e.g. **(2 marks)**.
- Where a calculation is required, show your working — marks are available for method as well as the final answer.
- Once you finish, check your answers against the mark scheme at the back of this document.
- This mock test is provided free by BrainyEdHub to give you a feel for our teaching standard — book a free trial class to go through it with a tutor.

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Question Paper

Computational Thinking

1. Define 'algorithm'. [1 mark]
2. Define 'decomposition' in computational thinking. [1 mark]
3. Define 'abstraction' in computational thinking. [1 mark]
4. State one benefit of using a flowchart to design an algorithm. [1 mark]
5. Name a common way of representing an algorithm other than a flowchart. [1 mark]

Data Representation

6. State the two digits used in the binary number system. [1 mark]
7. Convert the denary number 13 to binary. [2 marks]
8. Convert the binary number 1010 to denary. [2 marks]
9. State the term for a single binary digit. [1 mark]
10. State how many bits make up one byte. [1 mark]
11. Convert the denary number 20 to binary. [2 marks]
12. State one reason images and sound must be converted to binary for computer storage. [1 mark]

Programming

13. Define 'variable' in the context of programming. [1 mark]
14. State the data type used to store a whole number. [1 mark]
15. State the data type used to store a True/False value. [1 mark]
16. Explain what an IF...ELSE statement is used for in a program. [2 marks]
17. Define 'loop' and give one example of a type of loop. [2 marks]
18. State the data type used to store text such as a name. [1 mark]
19. Define 'syntax error' in programming. [1 mark]
20. State the purpose of a 'comment' in program code. [1 mark]

Networks & Security

- 21.** What does CPU stand for? **[1 mark]**
- 22.** State one function of RAM in a computer. **[1 mark]**
- 23.** What does LAN stand for? **[1 mark]**
- 24.** State one difference between a LAN and a WAN. **[2 marks]**
- 25.** Define 'cyber security' and give one example of a common threat. **[2 marks]**
- 26.** Define 'malware'. **[1 mark]**
- 27.** State one way a user can help keep their online accounts secure. **[1 mark]**

Mark Scheme

Use this guide to check your answers. Award full marks for correct final answers; where working is shown, award partial credit as indicated.

Q	Model answer / mark scheme	Marks
1	A precise, step-by-step set of instructions for solving a problem or completing a task.	1
2	Breaking a complex problem down into smaller, more manageable sub-problems.	1
3	Removing unnecessary detail from a problem so you can focus on what's important.	1
4	E.g. it makes the logic easy to visualise and follow before writing any code.	1
5	Pseudocode	1
6	0 and 1	1
7	1101 $8 + 4 + 1 = 13.$	2
8	10 $8 + 0 + 2 + 0 = 10.$	2
9	A bit	1
10	8	1
11	10100 $16 + 4 = 20.$	2
12	Computers can only store and process data as binary (electronic on/off signals).	1
13	A named location in memory used to store a value that can change while a program runs.	1
14	Integer	1
15	Boolean	1
16	It allows a program to make a decision — running one block of code if a condition is true, and a different block if it is false.	2
17	A structure that repeats a set of instructions; e.g. a FOR loop or a WHILE loop.	2
18	String	1
19	An error caused by code that does not follow the rules of the programming language.	1
20	To explain what the code does for anyone reading it, without affecting how the program runs.	1
21	Central Processing Unit	1

Q	Model answer / mark scheme	Marks
22	It temporarily stores data and instructions currently being used by running programs.	1
23	Local Area Network	1
24	A LAN covers a small area (e.g. one building); a WAN covers a large geographical area, connecting multiple LANs (e.g. the internet).	2
25	The practice of protecting systems, networks and data from digital attack; e.g. malware, phishing, or a virus.	2
26	Software that is deliberately designed to damage a computer system or gain unauthorised access to it.	1
27	E.g. using a strong, unique password, or enabling two-factor authentication.	1
Total for Computer Science		34 marks

How did you do? A BrainyEdHub tutor can go through any question you got stuck on, and build a personalised revision plan around your results. Book a free trial class at brainyedhub.com/contact.html or message us on WhatsApp.